

- Q.27 Explain Ionic potential of body and how it is converted into electric current.
- Q.28 Explain Electrode used for measuring muscle signals.
- Q.29 Write a note on LVDT?
- Q.30 Explain Active Transducer?
- Q.31 Write a note on applications of OFS (Optical fiber Sensor)?
- Q.32 Explain two types of Blood Flow?
- Q.33 Write a note on preamplifier?
- Q.34 Explain instrument used for measuring heart sound.
- Q.35 Draw and explain block diagram of basic recorder system.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Why transducers and sensors are important for healthcare system?
- Q.37 Explain EMG, EEG and ECG electrodes.
- Q.38 Explain different type of Temperature transducer, principle of Thermistor and IR thermometer?

5th Sem / Med. Eltx.

Subject:- Biomedical Sensors and Transducers (BMST)

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 M-I System is also known as:
- Major instrument System
 - Monitor instrument System
 - Man-instrumentation System
 - None of these
- Q.2 Device for converting energy into another form is known as:
- Time
 - Tex
 - Transducer
 - none of these
- Q.3 Unit of Time is:
- Pixel
 - Second
 - Hertz
 - none of these
- Q.4 Property of Biomaterial is:
- Non-Toxic
 - Light Weight
 - BIO-Compatible
 - All of these

Q.5 Role of Electrode is?

- a) to transfer Data
- b) Transformer
- c) To pick signal from body
- d) none of these

Q.6 Two characteristics of Sensors are:

- a) Accuracy b) Precision
- c) Both d) none of these

Q.7 LVDT is a type of:

- a) Light transducer
- b) Differential Transformer
- c) Pixel Transducer
- d) none of these

Q.8 Biomedical electrode can be of:

- a) Suction Type b) Needle
- c) Surface electrodes d) all of these

Q.9 Body potential is generated due to:

- a) Non-metallic b) Non-ionic
- c) Ions d) none of these

Q.10 Muscle signals are also known as :

- a) Electroencephalography
- b) Electrocardiography
- c) Electromyography
- d) none of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Write two examples of measurement.

Q.12 Define Biomaterial.

Q.13 Define EMG.

Q.14 Write name of two types of electrodes?

Q.15 Write strain gauge formula?

Q.16 Define sensor.

Q.17 Write one property of smart sensor?

Q.18 Define Electromagnetic blood flow.

Q.19 Define Piezoelectric property.

Q.20 Define Biomedical recorders.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Write a note on Man-instrumentation System and its role in medical field?

Q.22 Explain difference between Instrumentation and Biomedical Instrumentation?

Q.23 Explain electrode and its role with human body.

Q.24 Classify Transducers.

Q.25 Write five property of Sensors?

Q.26 Explain Photoelectric Transducers and its two applications.